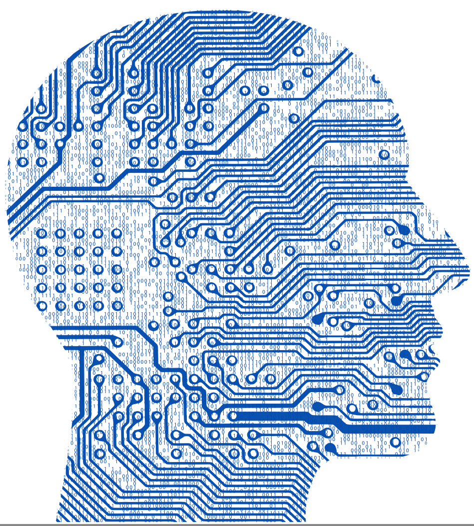
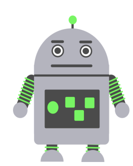
AI / ML COURSE

7 - Month Course Plan

BEGINNER TO JOB-READY



AI / ML COURSE



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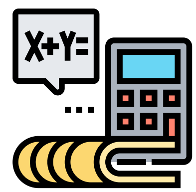
🧠 AI / ML Engineer – Complete Syllabus Roadmap

🔹 PART 1: AI/ML FOUNDATION COURSE (2 Months – 8 Weeks)

Week 1 – Mathematical Foundations

- Linear Algebra: vectors, matrices, dot product, eigen concepts
- Probability basics: events, distributions
- Statistics: mean, variance, standard deviation
- Calculus basics: derivatives, gradients (intuition only)



Week 2 – Python for Data Science

- Python fundamentals (loops, functions, OOP basics)
- NumPy (arrays, broadcasting, vectorization)
- Pandas (dataframes, indexing, merging)
- Jupyter Notebook workflow

Week 3 – Data Handling & EDA

- Data cleaning (missing values, duplicates)
- Data preprocessing & scaling
- Feature engineering basics

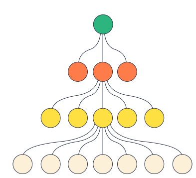
- Exploratory Data Analysis (EDA)
- Data visualization (Matplotlib, Seaborn)

Week 4 – Supervised Learning (Regression)

- Machine Learning pipeline
- Linear Regression
- Multiple Linear Regression
- Logistic Regression
- Bias–variance tradeoff
- Hands-on mini project

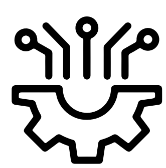
Week 5 – Supervised Learning (Classification)

- Decision Trees
- Random Forest
- K-Nearest Neighbors (KNN)
- Support Vector Machines (SVM)
- Evaluation metrics (Accuracy, Precision, Recall, F1)

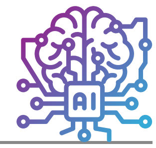


Week 6 – Unsupervised Learning

- K-Means clustering
- Hierarchical clustering

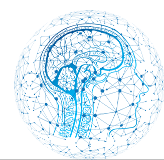


- Dimensionality reduction (PCA)
- Use cases & visualization



Week 7 – Deep Learning Basics

- Neural Network fundamentals
- Activation functions
- Backpropagation intuition
- CNN basics (image understanding)
- RNN basics (sequence data)

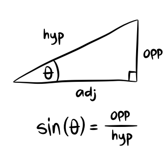


Week 8 – Model Evaluation & Deployment

- Cross-validation
- Hyper-parameter tuning
- Regularization techniques
- Intro to deployment
- Deploy ML model using **Streamlit**
- Foundation final project

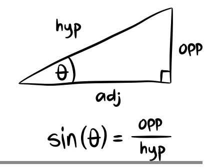
PART 2: ADVANCED AI/ML ENGINEER COURSE (5 Months – 20 Weeks)

Month 1: Advanced Math & ML (Weeks 9–12)



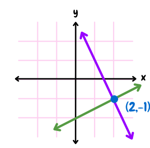
Week 9 – Advanced Mathematics

- Multivariate calculus
- Optimization techniques (GD, SGD)
- Convex optimization



Week 10 – Bayesian & Probabilistic ML

- Bayesian inference
- Probability models
- Probabilistic graphical models



Week 11 – Advanced Machine Learning

- Regularization (L1, L2, ElasticNet)
- Kernel methods
- Model selection strategies

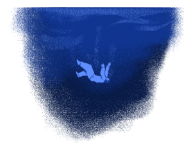
Week 12 – Ensemble Learning

- Gradient Boosting
- XGBoost
- LightGBM
- Feature importance & tuning

Month 2: Deep Learning Specialization (Weeks 13–16)

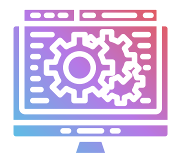
Week 13 – CNN Deep Dive

- CNN architectures
- Image classification
- Object detection basics
- Transfer learning



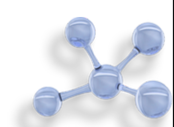
Week 14 – Advanced Computer Vision

- Object detection (YOLO, SSD, Faster R-CNN)
- Image segmentation (U-Net, Mask R-CNN)



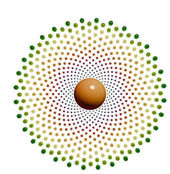
Week 15 – RNNs & Transformers

- RNN, LSTM, GRU
- Seq2Seq models
- Attention mechanism
- Transformers overview



Week 16 – Generative Models

- GANs
- VAEs

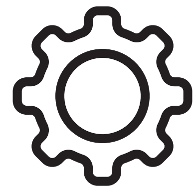


- Diffusion models
- Image & text generation projects

Month 3: NLP & LLMs (Weeks 17–20)

Week 17 – NLP Fundamentals

- Text preprocessing
- Tokenization
- Word embeddings (Word2Vec, GloVe, FastText)



Week 18 – Advanced NLP Tasks

- Sentiment analysis
- Named Entity Recognition (NER)
- Text summarization
- Question Answering

Week 19 – Large Language Models

- Transformer architecture deep dive
- BERT, GPT overview
- Prompt engineering
- Fine-tuning LLMs

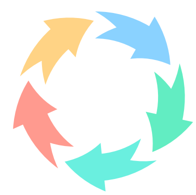
Week 20 – LLM Applications

- Chatbots
- RAG systems
- LLM evaluation
- LLM deployment basics

Month 4: MLOps & Production (Weeks 21–24)

Week 21 – MLOps Fundamentals

- ML lifecycle
- Model versioning (DVC, MLflow)
- Experiment tracking



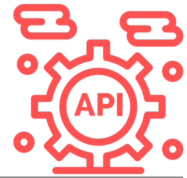
Week 22 – CI/CD for ML

- Git & GitHub
- CI/CD pipelines
- GitHub Actions / Jenkins for ML

Week 23 – Deployment & Scaling

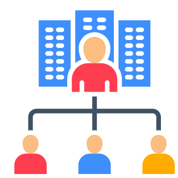
- Docker

- Kubernetes basics
- REST APIs for ML
- Scaling inference



Week 24 – Monitoring & Reliability

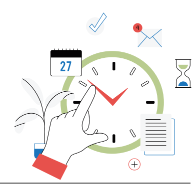
- Model monitoring
- Data & concept drift
- Logging & alerting
- Production best practices



📌 Month 5: Special Topics & Capstone (Weeks 25–28)

Week 25 – Advanced Topics

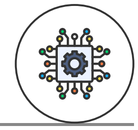
- Graph Neural Networks (GNNs)
- Time-Series forecasting
- Anomaly detection



Week 26 – Ethical & Responsible AI

- Bias & fairness
- Explainable AI
- Privacy-preserving ML

- Federated learning & DP



Week 27 – Capstone Project (Build Phase)

- Problem definition
- Data pipeline
- Model development
- Evaluation

Week 28 – Capstone Project (Deploy & Present)

- End-to-end deployment
- Documentation
- Presentation & demo